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HUJIE WANG

3A Computer Science

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LANGUAGES AND TECHNOLOGIES

Languages: C++; C; Java; Python; C#; SQL; JavaScript; Octave; MATLAB; Lua; Scheme; Pascal; XML
IDEs: Visual Studio; Eclipse; Pycharm; Emacs; Vim
Libraries: LIBSVM, Link Grammar, Stanford NLP Parser
Version Control Systems: Git; SVN; CVS

EMPLOYMENT

Software Engineer, Intern | **Evertz Microsystem Ltd.** | **May.2014-Aug.2014**

- Implemented a video compression analysis tool which provides the user with a visual representation of the encoded video features of AVC/H.264 streams (Python)
- Modified H.264 and MPEG-2 decoder for encoding algorithm analysis by implementing a custom decoded picture buffer and a command line interface for writing encoding parameters into files (C and C++)
- Created an automation script to configure and start FPGA encoders, capture and decode encoded stream with FFmpeg and evaluate the video quality by comparing Peak signal-to-noise ratio data (Python, C++ and Bash)

Software Engineer, Intern | **AppZero Software Corp.** | **Sept.2013-Dec.2013**

- Implemented a function to detect disconnections between the local machine and the source machine (C#, C and C++)
- Improved the GUI Interface of AppZero Tether Console and the Administration Console (C#, C and C++)

EDUCATION

Waterloo, Ontario | **University of Waterloo** | **Fall 2012 – Fall 2016 (Expected)**

- Candidate for Bachelor of Computer Science, Co-operative Program
- Relevant Courses: Operating Systems; Databases; Algorithms; Computer Organization and Design; Data Structures; Introduction to Combinatorics; Foundation of Sequential Programs; Statistics
- University of Waterloo President's Scholarship of Distinction, University of Waterloo, Waterloo, ON (2012)

PROJECTS

Anaphora Resolution-Google Summer of Code (2014) (Python and Scheme)
github.com/opencog/opencog/tree/master/opencog/nlp/anaphora

- Implemented a modified Hobbs Algorithm in a MindAgent for Anaphora Resolutions and Zero-Pronoun Resolutions on Dependency Parse Trees
- Developed rule-based sieves to eliminate unqualified anaphora antecedents with nearly 100% accuracy

Spam Classification (2014) (Octave)
github.com/hujiewang/SpamClassification

- Implemented a Support Vector Machine with Linear Kernel for spam classification
- Achieved 98.8% accuracy in the testing data set of 1000 test examples by training 4000 examples of spam and non-spam email

Handwritten Digits Recognition (2014) (Octave)
github.com/hujiewang/HandwrittenDigitsRecognition

- Implemented a Neural Network with three layers for handwritten digits recognition
- Achieved 97.5% accuracy by training 5000 examples of 20 pixel by 20 pixel grayscale image of digit

SimpleSaving Android Project (2013) (Java, XML)
github.com/hujiewang/SimpleSaving

- Designed and implemented an Android application to provide an intuitive way of budge planning

Tetris (2013) (C++)

- Implemented a Tetris game as the final project of Object-Oriented Software Development Course

ADDITIONAL EXPERIENCE AND AWARDS

Programming Contests: 17th in 2014 ACM ICPC North America Qualifier-University of Waterloo; 10th in Spring 2014 Waterloo ACM local contest; Word ranking 774 on UVa online judge; Competed in Topcoder, CodeForces, Google Code Jam, Yandex.Algorithm, Zepto Code Rush and MemSQL Start[c]UP 2.0
Online Courses: Coursera Stanford Machine Learning course; Udacity HTML5 Game Development course